

SHAURYA AGRAWAL

EDUCATION

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Manipal University Jaipur

Jaipur, Rajasthan

Bachelor of Technology in Computer Science Engineering

Expected May 2027

- **Relevant Coursework:** Data Structures & Algorithms, Deep Learning, Computer Architecture & Concurrency, Embedded System Design
- **IEEE Student Branch:** Coordinated logistics for 5+ hackathons, growing department-wide participation.

EXPERIENCE

AI Systems Engineer Intern

Jun 2025 – Aug 2025

Sarvaha Systems

On-Site

- Built and validated an enterprise AI customer-support platform (FastAPI/NestJS/React microservices, Agno multi-agent orchestration), writing pytest unit/integration suites wired into a Jenkins CI pipeline and designing test scenarios for multi-turn agent conversations to catch persistent-memory edge cases.
- Designed a RAG pipeline over Milvus and sentence-transformer embeddings, benchmarking chunking and re-ranking strategies against a retrieval-accuracy test set spanning OCR'd PDFs, web scrapes, and video transcripts.
- Built an asynchronous ingestion system (Celery/RabbitMQ, SSE + Redis Pub/Sub) and load-tested the streaming layer to verify response-latency targets for real-time AI answers.
- Instrumented full-stack observability (Langfuse, OpenTelemetry, CloudWatch) to monitor production behavior and triage regressions, and verified RBAC/multi-tenant isolation before deployment via Docker behind a Traefik reverse proxy.

Cybersecurity Operations Intern

Jun 2024 – Aug 2024

DCDIUM Technologies

On-Site

- Investigated 20-50 weekly security alerts as part of the Blue Team, triaging and documenting phishing, DDoS, and APT activity across enterprise infrastructure.
- Executed vulnerability assessments on 15+ systems via Nessus/OpenVAS, identifying and documenting high-risk CVEs and driving remediation that cut attack surface by 40%.
- Authored 8 incident-response playbooks and led SIEM log analysis in Splunk, improving threat detection accuracy to 95% while communicating issue impact to engineering stakeholders.

PROJECTS

GraphRAG Security Reasoning Engine | Python, Neo4j, GraphRAG, Pydantic, MITRE ATT&CK

- Engineered a 3-phase GraphRAG pipeline (Ingestion → Retrieval → Generation) parsing 5 log formats (Syslog, CEF, CloudTrail, K8s Audit) into a Neo4j graph (9-node, 13-relationship ontology) and validated multi-hop attack-chain reconstruction against known incident scenarios.
- Built a hybrid retrieval system (vector search + bounded 4-hop graph traversal) with confidence-tiered filtering (entity ≥ 0.7 , relationship ≥ 0.8), validated via a pytest suite run in GitHub Actions CI to minimize false-positive connections in the graph.
- Integrated 3 threat-intel APIs (NVD CVE/CVSS, MITRE ATT&CK STIX, OTX AlienVault) for citation-grounded, auditable reasoning across lateral movement and CVE-exploitation chains.

AutoStream Social-to-Lead Agent | Python, LangGraph, LangChain, FAISS, Redis

- Built a stateful conversational agent in LangGraph/LangChain with persistent multi-turn memory, writing test cases across 3 distinct intent-routing paths to catch state-handling regressions.
- Designed and load-tested a RAG pipeline via FAISS vector search with tuned chunking (400 chars/60 overlap) to resolve product queries with high accuracy.
- Developed an automated lead-qualification state machine with strict Python-level gating to prevent LLM hallucinations, integrated with WhatsApp Business API via FastAPI and Redis session storage (24-hr TTL).

HarmonicAI | Python, Scikit-Learn, Streamlit, GenAI

- Built a hybrid ML audio recommendation engine (unsupervised intent clustering + Random Forest over 7+ acoustic features) and stress-tested its 3-layer NLP safety system (regex, TF-IDF, semantic routing) to verify 100% real-time content moderation.
- Built a GenAI Streamlit pipeline generating personalized therapy scripts across 5 psychological intents, securing it with PII salting, audit logging, and rate-limit testing (30 req/min).

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, JavaScript, Bash

Testing & QE: Test Case Design, pytest, CI/CD (Jenkins, GitHub Actions), Debugging & Root-Cause Analysis, Vulnerability Assessment (Nessus, OpenVAS), Observability (Splunk, Langfuse, OpenTelemetry)

Frameworks & AI: FastAPI, NestJS, React, LangGraph, LangChain, PyTorch, TensorFlow, Scikit-learn, Pandas, Streamlit

Infra & Tools: Linux, Git, Docker, PostgreSQL, Neo4j, Milvus, FAISS, Redis, RabbitMQ, Celery, AWS (S3, EC2), SQLAlchemy, Prisma

Core: System Design, Microservices, Multi-Agent Systems, RAG, Knowledge Graphs, ML, Real-time Anomaly Detection, DSA

CERTIFICATIONS

- **Google Cybersecurity Professional Certificate** – Google (2025) | Credential Link
- **Supervised Machine Learning Certificate** – DeepLearning.AI, Stanford University (2024)